

Resistances

Classification

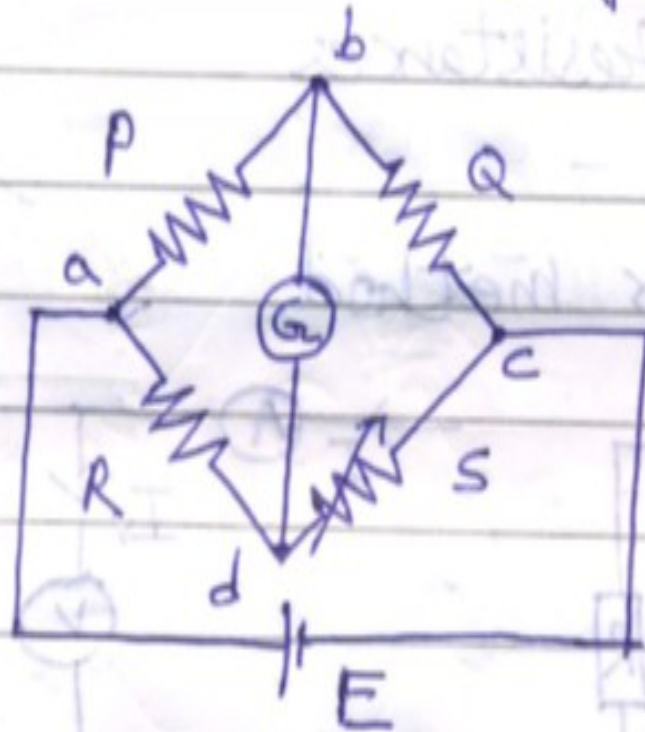
- (i) Low resistances ($\leq 1\Omega$)
- (ii) Medium resistances ($1\leq R\leq 0.1\text{ M}\Omega$)
- (iii) High resistances ($0.1\text{ M}\Omega$ and above)

Measurement of Medium Resistance

Wheatstone Bridge

WSB

(Highly accurate)

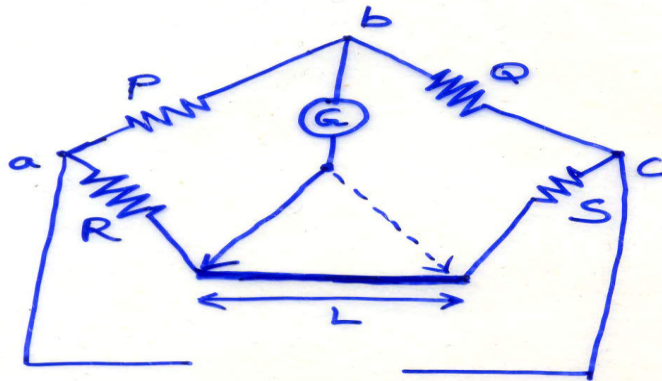


PQ → ratio arms

S → Standard arm

$$R = \frac{PS}{Q}$$

Carey Foster Slide Wire Bridge



For first balance

$$\frac{P}{Q} = \frac{R + l_1 \sigma}{S + (L - l_1) \sigma} \dots (i)$$

$\sigma \rightarrow$ resistance per unit length

For second balance

$$\frac{P}{Q} = \frac{S + l_2 \sigma}{R + (L - l_2) \sigma} \dots (ii)$$

Equating $\frac{P}{Q} + 1$ on both sides

$$\frac{R + S + L \sigma}{S + (L - l_1) \sigma} = \frac{S + R + L \sigma}{R + (L - l_2) \sigma}$$

$$\Rightarrow S-R = (l_1 - l_2) \times \dots \text{ (iii)}$$

($\times$) is eliminated by shunting S with a known resistance.

Let balance achieved after shunting S' be at l'_1 & l'_2 .

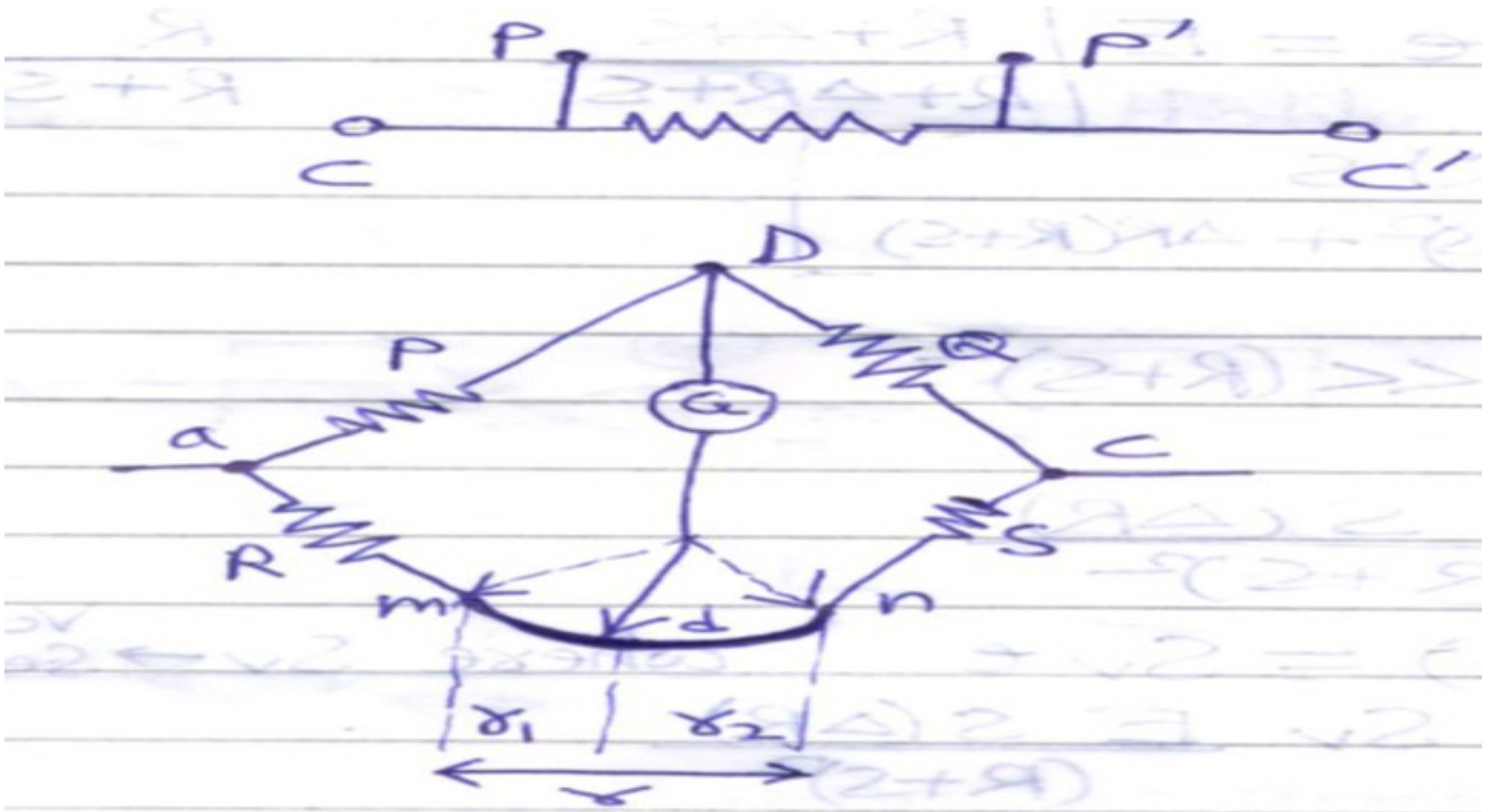
$$\Rightarrow S' - R = (l'_1 - l'_2) \times \dots \text{ (iv)}$$

$$\Rightarrow \frac{S-R}{l_1 - l_2} = \frac{S'-R}{l'_1 - l'_2}$$

$$\Rightarrow R = \frac{S(l'_1 - l'_2) - S'(l_1 - l_2)}{[(l'_1 - l'_2) - (l_1 - l_2)]}$$

Measurement of Low Resistance

Principle of Kelvin's Bridge



At balance

$$\frac{P}{Q} = \frac{R + \gamma_1}{S + \gamma_2}$$

$$\Rightarrow R + \gamma_1 = \frac{P}{Q} (S + \gamma_2) \quad \dots (i)$$

$$\Rightarrow \text{If } \frac{\gamma_1}{\gamma_2} = \frac{P}{Q}$$

$$\Rightarrow \frac{\gamma_1}{\gamma_1 + \gamma_2} = \frac{P}{P + Q}$$

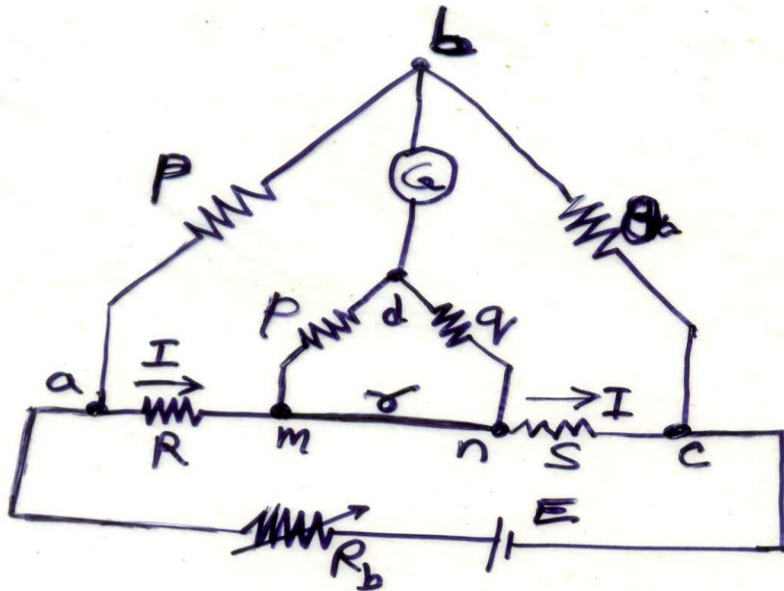
$$\Rightarrow \gamma_1 = \frac{P}{P + Q} \gamma$$

Similarly $\delta_2 = \frac{Q}{P+Q} \delta$

$$\Rightarrow R + \left(\frac{P}{P+Q} \right) \delta = \frac{P}{Q} \left(S + \frac{Q}{P+Q} \delta \right) \quad \text{From (1)}$$

$$\Rightarrow R = \frac{P}{Q} S$$

Kelvin Double Bridge



'R' extends between am
'S' extends between nc.

$\delta \rightarrow$ resistance of lead
that connects R
with S.

It incorporates a
second set of
ratio arms.

$$\frac{P}{Q} = \frac{p}{q} \quad \& \quad \text{At balance } E_{ab} = E_{amd}$$

$$\Rightarrow E_{ab} = \frac{P}{P+Q} E_{ac} \quad \& \quad E_{ac} = I \left[R + S + \frac{(P+Q)\delta}{P+Q+\delta} \right]$$

$$E_{amd} = I \left[R + \frac{p}{p+q} \left\{ \frac{(p+q)r}{p+q+r} \right\} \right] = I \left[R + \frac{pr}{p+q+r} \right]$$

for zero galvanometer deflection

$$E_{ab} = E_{amd}$$

$$\Rightarrow \frac{p}{p+q} I \left[R + s + \frac{(p+q)r}{p+q+r} \right] = I \left[R + \frac{pr}{p+q+r} \right]$$

$$\Rightarrow R = \frac{p}{q} s + \frac{qr}{p+q+r} \left[\frac{p}{q} - \frac{p}{q} \right]$$

$$\text{If } \frac{p}{q} = \frac{p}{q}, \quad R = \frac{p}{q} s$$

Measurement of High Resistance

Three Terminal Resistance

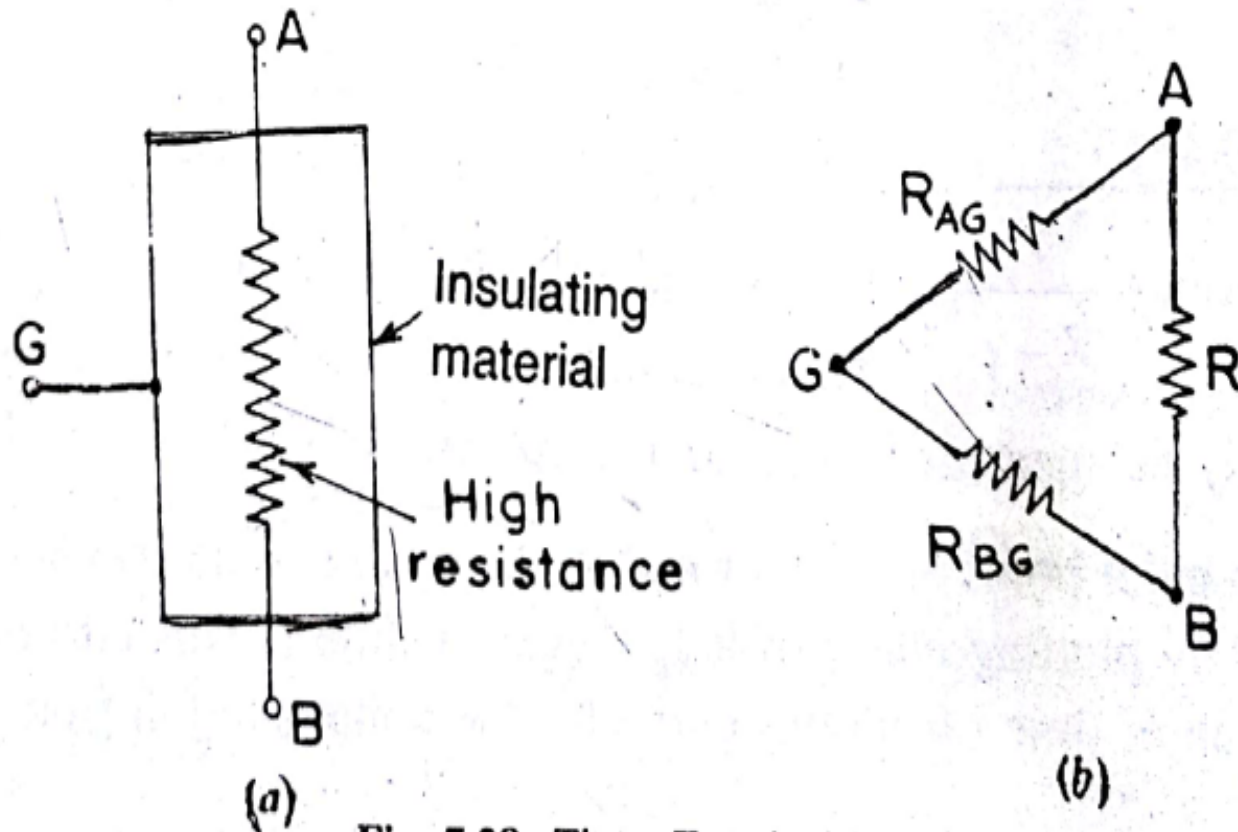
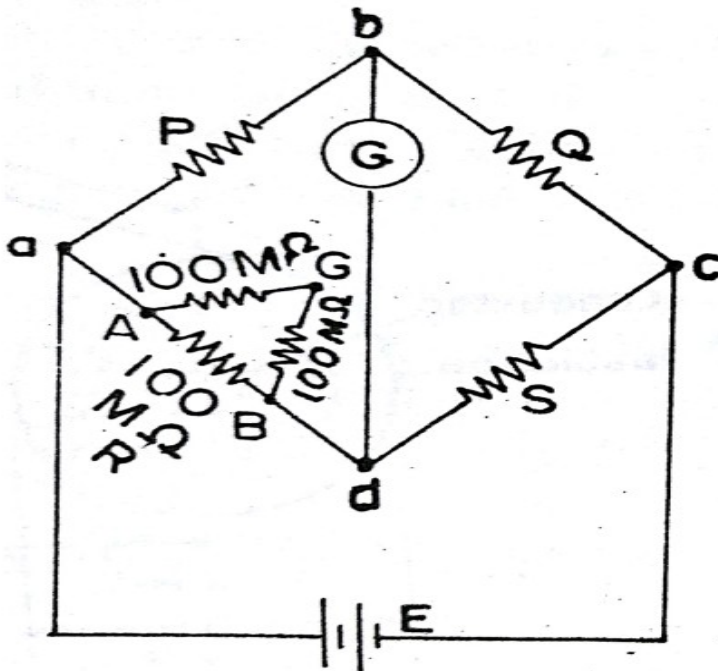


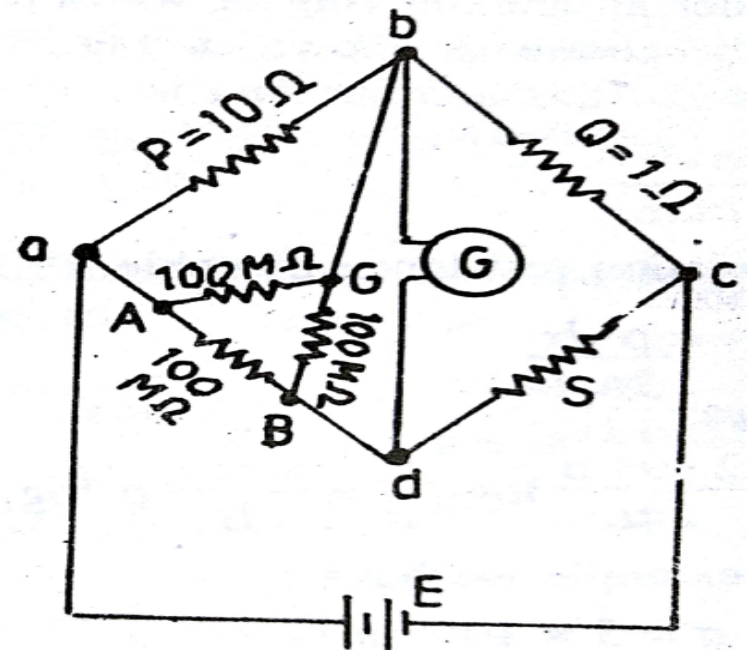
Fig. 7.28. Three Terminal Resistances.

Use of Guard Circuit

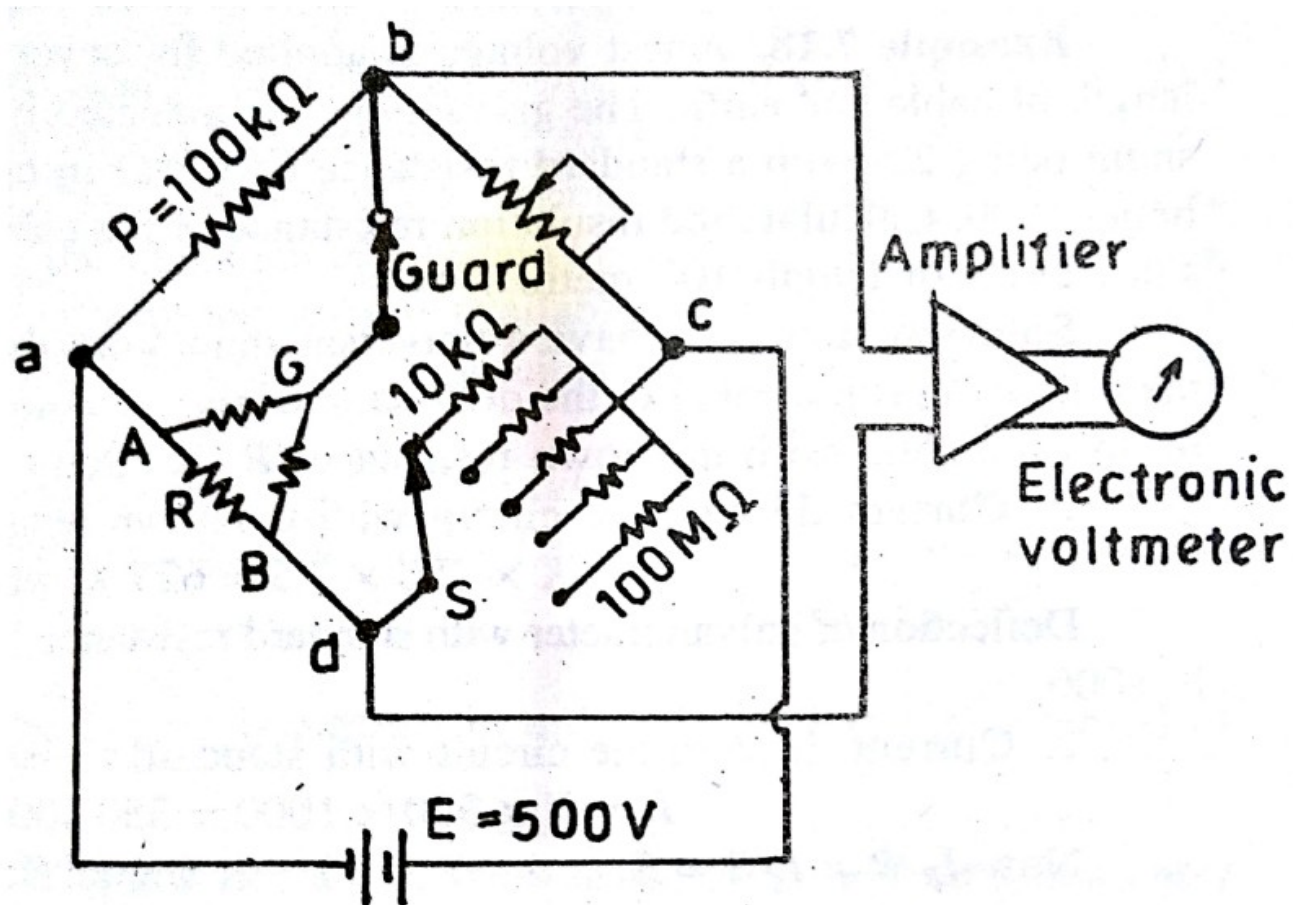
Without Guard (Error)



With Guard



Megaohm Bridge



Megger

